## ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 21 | Sum $= 105$; $\div 5$ |
| 2 | **(c)** 28 | Arrange in order; take the middle value |
| 3 | **(a)** 18 | Most frequent value |
| 4 | **(c)** 33 | Highest − lowest |
| 5 | **(c)** 90 | Sum = mean × number of values |
| 6 | **(b)** one observation | One dot per data value |
| 7 | **(a)** 13 | Sum $= 65$; $\div 5$ |
| 8 | **(b)** 24 | Arrange in order; take the middle value |
| 9 | **(d)** 7 | Most frequent value |
| 10 | **(b)** 44 | Highest − lowest |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Sum $= 140$ (1); mean $= 140 \div 5 = 28$ (1). | 1+1 |
| 12 | Ascending order: 5, 32, 33, 38, 53, 56, 60 (1); median $= 38$ (1). | 1+1 |
| 13 | Range $= 15 - 9 = 6$ (1); mode $= 10$ (1). | 1+1 |
| 14 | Sum $= 190$ (1); mean $= 190 \div 5 = 38$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Mean $= 270 \div 6 = 45$ (2); students above mean $= 2$ (1). | 2+1 |
| 16 | Ascending: 6, 7, 21, 37, 44, 50, 55, 69, 70 (1); 9 values, middle one is the 5th (1); median $= 44$ (1). | 1+1+1 |
| 17 | Rises: 4, 7, 3 (2); greatest rise 7°C (1). | 2+1 |
| | **OR:** Mean $= 318 \div 6 = 53$ (2); students above mean $= 3$ (1). | 2+1 |
| 18 | Ascending: 21, 28, 29, 45, 47, 54, 55, 58, 65 (1); 9 values, middle one is the 5th (1); median $= 47$ (1). | 1+1+1 |
| | **OR:** Rises: 3, 9, 3 (2); greatest rise 9°C (1). | 2+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Mean $= 15.17$ (1); median $= 15.5$ (2); mode $= 25$ (1); range $= 21$ (1). | 1+2+1+1 |
| | **OR:** Sum of five $= 5 \times 20 = 100$ (2); sum of four $= 85$ (1); fifth $= 15$ (2). | 2+1+2 |
| 20 | Mean $= 13.5$ (1); median $= 10.5$ (2); mode $= 7$ (1); range $= 21$ (1). | 1+2+1+1 |
| | **OR:** Sum of five $= 5 \times 15 = 75$ (2); sum of four $= 55$ (1); fifth $= 20$ (2). | 2+1+2 |

**Mistakes to avoid:**

- Finding the median without arranging the data in order first.
- For an even number of observations, taking one of the middle values instead of the **mean of the two middle values**.
- Confusing mode with median or mean: mode is the most frequent value, not the middle value.
- Dividing the sum by the number of *distinct* values instead of the total number of observations (counting repeated values only once).
- Computing range as the difference between the first and last values as listed, not the maximum and minimum.
- Reading a line graph at the wrong point or ignoring the scale of the axes.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 96 | Sum = mean × number of values |
| 2 | **(b)** mode | Definition of mode |
| 3 | **(c)** 17 | Sum $= 85$; $\div 5$ |
| 4 | **(a)** 27 | Arrange in order; take the middle value |
| 5 | **(b)** 8 | Most frequent value |
| 6 | **(b)** 55 | Highest − lowest |
| 7 | **(d)** 250 | Sum = mean × number of values |
| 8 | **(c)** range | Definition of range |
| 9 | **(d)** 14 | Sum $= 70$; $\div 5$ |
| 10 | **(b)** 25 | Arrange in order; take the middle value |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Ascending order: 7, 10, 15, 23, 27, 41, 56 (1); median $= 23$ (1). | 1+1 |
| 12 | Range $= 24 - 6 = 18$ (1); mode $= 8$ (1). | 1+1 |
| 13 | Sum $= 155$ (1); mean $= 155 \div 5 = 31$ (1). | 1+1 |
| 14 | Ascending order: 16, 24, 31, 45, 52, 56, 59 (1); median $= 45$ (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Mean $= 396 \div 6 = 66$ (2); students above mean $= 3$ (1). | 2+1 |
| 16 | Ascending: 6, 12, 18, 23, 32, 34, 44, 60, 62 (1); 9 values, middle one is the 5th (1); median $= 32$ (1). | 1+1+1 |
| 17 | Rises: 3, 3, 9 (2); greatest rise 9°C (1). | 2+1 |
| | **OR:** Mean $= 294 \div 6 = 49$ (2); students above mean $= 2$ (1). | 2+1 |
| 18 | Ascending: 6, 16, 22, 27, 30, 54, 64, 66, 70 (1); 9 values, middle one is the 5th (1); median $= 30$ (1). | 1+1+1 |
| | **OR:** Rises: 7, 4, 3 (2); greatest rise 7°C (1). | 2+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Mean $= 15.83$ (1); median $= 14$ (2); mode $= 6$ (1); range $= 23$ (1). | 1+2+1+1 |
| | **OR:** Sum of five $= 5 \times 30 = 150$ (2); sum of four $= 116$ (1); fifth $= 34$ (2). | 2+1+2 |
| 20 | Mean $= 14.33$ (1); median $= 11$ (2); mode $= 27$ (1); range $= 23$ (1). | 1+2+1+1 |
| | **OR:** Sum of five $= 5 \times 12 = 60$ (2); sum of four $= 48$ (1); fifth $= 12$ (2). | 2+1+2 |

**Mistakes to avoid:**

- Finding the median without arranging the data in order first.
- For an even number of observations, taking one of the middle values instead of the **mean of the two middle values**.
- Confusing mode with median or mean: mode is the most frequent value, not the middle value.
- Dividing the sum by the number of *distinct* values instead of the total number of observations (counting repeated values only once).
- Computing range as the difference between the first and last values as listed, not the maximum and minimum.
- Reading a line graph at the wrong point or ignoring the scale of the axes.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Tales by Dots and Lines** | Class 8 Maths | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 2 | Most frequent value |
| 2 | **(c)** 29 | Highest − lowest |
| 3 | **(d)** 126 | Sum = mean × number of values |
| 4 | **(b)** change over time | Trend |
| 5 | **(a)** 17 | Sum $= 85$; $\div 5$ |
| 6 | **(d)** 25 | Arrange in order; take the middle value |
| 7 | **(b)** 7 | Most frequent value |
| 8 | **(b)** 52 | Highest − lowest |
| 9 | **(b)** 126 | Sum = mean × number of values |
| 10 | **(b)** 7 | Mean of equal values |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Range $= 23 - 3 = 20$ (1); mode $= 5$ (1). | 1+1 |
| 12 | Sum $= 135$ (1); mean $= 135 \div 5 = 27$ (1). | 1+1 |
| 13 | Ascending order: 7, 30, 45, 48, 49, 55, 60 (1); median $= 48$ (1). | 1+1 |
| 14 | Range $= 25 - 4 = 21$ (1); mode $= 11$ (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Mean $= 372 \div 6 = 62$ (2); students above mean $= 3$ (1). | 2+1 |
| 16 | Ascending: 6, 9, 14, 19, 34, 36, 52, 62, 69 (1); 9 values, middle one is the 5th (1); median $= 34$ (1). | 1+1+1 |
| 17 | Rises: 2, 9, 4 (2); greatest rise 9°C (1). | 2+1 |
| | **OR:** Mean $= 402 \div 6 = 67$ (2); students above mean $= 4$ (1). | 2+1 |
| 18 | Ascending: 12, 15, 30, 36, 38, 62, 68, 69, 70 (1); 9 values, middle one is the 5th (1); median $= 38$ (1). | 1+1+1 |
| | **OR:** Rises: 2, 7, 10 (2); greatest rise 10°C (1). | 2+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Mean $= 19.5$ (1); median $= 21.5$ (2); mode $= 23$ (1); range $= 13$ (1). | 1+2+1+1 |
| | **OR:** Sum of five $= 5 \times 45 = 225$ (2); sum of four $= 180$ (1); fifth $= 45$ (2). | 2+1+2 |
| 20 | Mean $= 16$ (1); median $= 16.5$ (2); mode $= 22$ (1); range $= 24$ (1). | 1+2+1+1 |
| | **OR:** Sum of five $= 5 \times 24 = 120$ (2); sum of four $= 94$ (1); fifth $= 26$ (2). | 2+1+2 |

**Mistakes to avoid:**

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- Confusing mode with median or mean: mode is the most frequent value, not the middle value.
- Dividing the sum by the number of *distinct* values instead of the total number of observations (counting repeated values only once).
- Computing range as the difference between the first and last values as listed, not the maximum and minimum.
- Reading a line graph at the wrong point or ignoring the scale of the axes.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.